

Dokažte, že v libovolném komplexním prostoru V se skalarův součin $\langle, \rangle$ plní
 (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\begin{aligned}
 \|u+v\|^2 - \|u-v\|^2 &= \langle u, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle u, u \rangle - \langle v, v \rangle - \\
 &\quad - \langle u, -v \rangle - \langle -v, u \rangle = \langle u, v \rangle + \overline{\langle u, v \rangle} + \langle u, v \rangle + \overline{\langle u, v \rangle} =
 \end{aligned}$$

$$2z + \bar{z} = 2\operatorname{Re}(z)$$

$$= 4\operatorname{Re}(\langle u, v \rangle)$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\begin{aligned}
 \|u-iv\|^2 - \|u+iv\|^2 &= \langle u, u \rangle + \langle -iv, -iv \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle - \\
 &\quad - \langle u, u \rangle - \langle iv, iv \rangle - \langle u, iv \rangle - \langle iv, u \rangle = -i\langle u, v \rangle + i\overline{\langle u, v \rangle} \\
 &\quad - i\langle u, v \rangle + i\overline{\langle u, v \rangle} = -i(\langle u, v \rangle - \overline{\langle u, v \rangle} + \langle u, v \rangle - \overline{\langle u, v \rangle})
 \end{aligned}$$

$$z = a+bi \quad z - \bar{z} = a+bi - a-bi = 2bi = 2i\operatorname{Im}(z)$$

$$\|u-iv\|^2 - \|u+iv\|^2 = -i(4i\operatorname{Im}(\langle u, v \rangle))$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$